

Ex. No: 2

Perform website, email, whois, DNS, and network footprinting on the target network

1. Aim

To conduct deep technical footprinting on a target network to extract infrastructural details including domain registration, DNS records, email routing configurations, and network topology.

2. Theoretical Background

Technical footprinting shifts from open-source web searching to querying specific infrastructural databases and protocols.

- **WHOIS:** A query and response protocol used for querying databases that store registered users or assignees of an Internet resource.
- **DNS Footprinting:** Extracting records (A, MX, NS, TXT) to understand how the target routes its traffic.
- **Email Footprinting:** Analyzing email headers to track the path an email took from sender to receiver.
- **Network Footprinting:** Mapping out the logical network topology using tools like traceroute.

3. Detailed Procedure

Step 1: WHOIS Lookup

Use a command-line utility or web interface to find domain registration details.

```
Command: whois rajalakshmi.org
```

Step 2: DNS Footprinting

Use `nslookup` or `dig` commands to query domain name servers and extract specific records.

```
Command: dig rajalakshmi.org ANY
Command: nslookup -type=MX rajalakshmi.org
```

Step 3: Network Footprinting (Traceroute)

Map the path packets take to reach the target domain.

```
Command (Linux): traceroute rajalakshmi.org
Command (Windows): tracert rajalakshmi.org
```

Step 4: Email Header Analysis

Obtain an email originating from the target domain, view the raw headers, and analyze the "Received:" lines to trace routing.

4. Observation & Outputs

4.1 WHOIS Output

```
Domain Name: RAJALAKSHMI.ORG
Registrar: GoDaddy.com, LLC
Creation Date: 1999-02-18T05:00:00Z
Registrar Registration Expiration Date: 2026-02-18T05:00:00Z
```

4.2 DNS Records

Record Type	Target / Data	Inference
A Record	104.21.x.x, 172.67.x.x	IPv4 Addresses resolving to the web server (Cloudflare routed).
MX Record	aspmx.l.google.com	Mail Exchange server for incoming emails (Google Workspace).
NS Record	ns1.rajalakshmi.org, ns2.rajalakshmi.org	Authoritative Name Servers managing the domain.

4.3 Network Trace (Traceroute)

```
Tracing route to rajalakshmi.org over a maximum of 30 hops:
  1    <1 ms    <1 ms    <1 ms    192.168.1.1
  2    12 ms    11 ms    14 ms    isp-gateway.net
  3    25 ms    24 ms    23 ms    core-router.chennai.net
  4    45 ms    44 ms    46 ms    cloudflare-gateway.net
Trace complete.
```

5. Result

Detailed technical footprinting successfully extracted the foundational architecture of the target network. By querying WHOIS, DNS, and mapping the network routes, a comprehensive overview of the target’s external-facing assets and routing paths was documented.